App Dev Project Report

1. Student Details

Name: Shri KRESHIV GULATI **Roll Number:** DS2400088 **Email:** 24f2000880@ds.study.iitm.ac.in

About Me: I am a student at IIT Madras BS Degree program with a deep interest in web application development and data-driven technologies. I enjoy building meaningful applications that combine learning, analytics, and user experience.

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2. Project Details

Project Title: Trek Management Application

Problem Statement:

Adventure organizations require efficient systems to manage trekking activities involving trek organizers, staff, and participants. Currently, many trekking groups rely on spreadsheets, phone calls, or manual coordination, which makes it difficult to manage trek approvals, track bookings, avoid overbooking, and maintain trek history.

Approach:

I first designed the database schema around the three user roles—Admin, Trek Staff, and Trekker—along with Trek and Booking entities. The application was organized using Flask Blueprints to separate authentication, admin, staff, and trekker functionalities. After implementing the core business logic, including role-based access, trek management, booking, and participant management, I built a responsive UI using Jinja2, Bootstrap, and a common `base.html` template for consistency.

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3. AI/LLM Declaration

I used **ChatGPT (GPT-5) and Claude(Sonnet 4.5)** to assist in writing SQLAlchemy model definitions, creating API documentation samples, and improving variable naming consistency.

The extent of AI/LLM usage is around **30–45%**, limited to **code suggestions and**

documentation formatting. All final implementation logic, debugging, and integration were done manually.

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4. Technologies and Frameworks Used

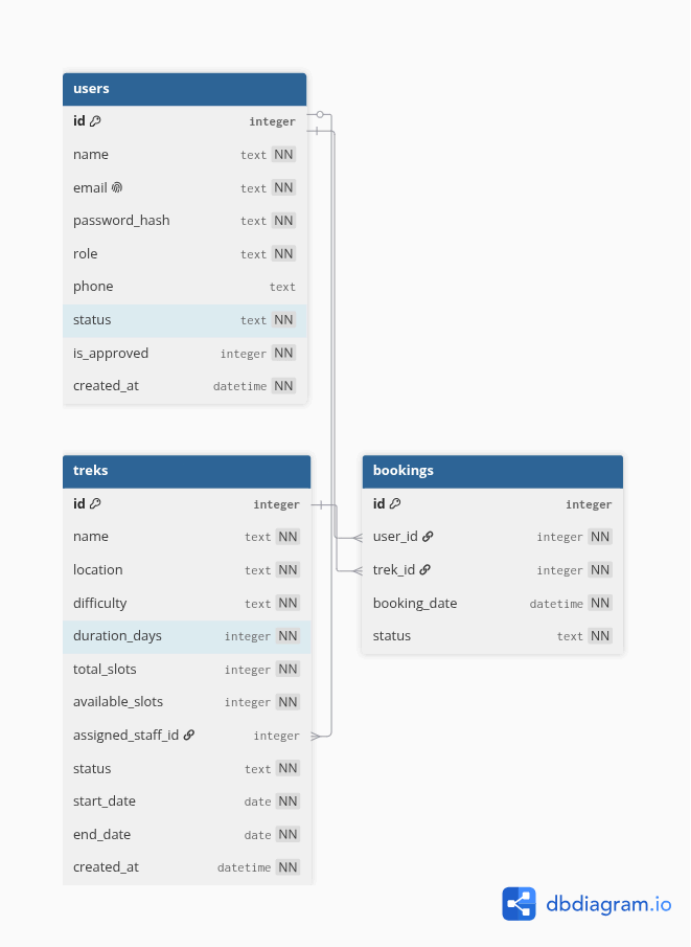
Technology / Library	Purpose
Flask	Core backend web framework
Flask-Login	User authentication, session management, and role-based login protection
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Werkzeug	Password hashing, secure filenames

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5. Database Schema / ER Diagram

3 tables, 3 relationships:

- users is the authentication table containing all users (admins, staff, and trekkers), distinguished by the **`role`** attribute.
- Staff users are assigned to treks through `assigned_staff_id` (one-to-many): one staff member can manage multiple treks, while each trek is assigned to at most one staff member.
- Users (trekkers) create bookings (one-to-many): one user can book multiple treks.
- Treks receive bookings (one-to-many): one trek can have multiple bookings.
- Bookings acts as the junction table between users and treks, implementing a many-to-many relationship. Each booking links one trekker to one trek and stores additional information such as booking date and booking status.



6. API Resource Endpoints

Endpoint	Method	Access	Description
`/api/treks`	GET	Admin, Staff, Trekker	Returns the list of treks. Admin can view all treks, while Staff and Trekkers can view only non-pending treks.
`/api/treks/<trek_id>`	GET	Admin, Staff, Trekker	Returns detailed information for a specific trek. Staff and Trekkers cannot access pending treks.

`/api/bookings`	GET	Admin, Staff, Trekker	Returns bookings based on user role. Admin sees all bookings, Staff sees bookings for assigned treks, and Trekkers see only their own bookings.
`/api/users`	GET	Admin	Returns the list of all registered users and staff (excluding the admin). Accessible only by the administrator.

7. Video Presentation

Drive Link:

https://drive.google.com/file/d/13dtF6shfHnj_54qQ6w8Dtmi1LRx0blMz/view?usp=sharing